सी एस आई आर – राष्ट्रीय भौतिक प्रयोगशाला
CSIR-NATIONAL PHYSICAL LABORATORY
(वैज्ञानिक तथा औद्योगिक अनुसंधान परिषद्)
(Council of Scientific and Industrial Research)
(राष्ट्रीय मापिकी संस्थान (एनएमआई), सदस्य बीआईपीएस एवं हस्ताक्षरकर्ता सीआईपीएम - एमआरए)
(National Metrology Institute (NMI), Member BIPM and Signatory CIPM - MRA)
डॉ. के. एस. कृष्णन् मार्ग, नई दिल्ली-110012, भारत
Dr. K. S. Krishnan Marg, New Delhi-110012 INDIA
दूरभाष/Phone : +91-11-4560, 8441, 8610, 9447, फैक्स/Fax : 91-11-4560 8448
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अंशांकन प्रमाण पत्र
CALIBRATION CERTIFICATE
WEIGHTS

प्रमाण पत्र संख्या / Certificate No.

25080617/D1.01/C-144

दिनांक/ Date	अगले अंशांकन हेतु अनुशंसित तिथि Recommended date for next calibration	पृष्ठ / Page	पृष्ठों की संख्या / No. of Pages
26.11.2025	26.11.2028	1	2

1. Calibrated for : HPCL Rajasthan Refinery Limited (HRRL)
Integrated Laboratory
Village: Sajiyali Roopji Kanthwada & Sambhra,
Tehsil : Pachpadra, District : Balotra
Rajasthan - 344032, India

Customer's Ref. No. : 31/WI/25-26, Dated : 29.07.2025
From : Weigh India, New Delhi

2. Description and identification of instrument

5 kg & 500 g (2 Nos.)	:	Integral knob cylindrical austenitic stainless steel weights.
1 mg (1 No.)	:	Austenitic stainless steel wire weight.
Assumed Density (d)	:	(7950 ± 140) kg/m³; $k = 2$
Make	:	Weigh India

3. Environmental conditions : Temperature : (23.0 ± 1.5) °C
Relative humidity : (50 ± 10) %
[Change in temperature during the calibration was less than ± 0.7 °C per hour]

4. Standard(s) used with associated Measurement uncertainty : NPL working standard (Sub-Div.#1.01/M/224(2)/00 & Sub-Div.#1.01/M/60027/55) o mass with uncertainty better than one-third of the reported uncertainty of measurement.

5. Metrological Traceability of Standard(s) used : The standard(s) used for calibration is(are) traceable to the National Standard which realize the units of quantities according to the International System of Units (SI).

6. Principle/Methodology of Calibration & Calibration Procedure number : CSIR-NPL Calibration Procedure No. Sub-Div.#1.01/Doc.3/ CP#WT/M-03. Method o comparison with the CSIR-NPL working standard(s) using substitution weighing. The reported mass value(s) is(are) the conventional mass value(s) (M_c) related to the true mass value(s) (M_t) by formula : $M_c = M_t \times [1 - 1.2 \times (1/d - 1/8\,000)]$.

अंशांकनकर्ता:
Calibrated by:
(D.C. SHARMA)

जाँचकर्ता:
Checked by:
(NIDHI SINGH)

प्रभारी वैज्ञानिक:
Scientist-in-charge:
(NIDHI SINGH)

जारीकर्ता:
Issued by:
डॉ. श्रीनिवास राव रागम
Dr. Srinivasa Rao Ragam

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CALIBRATION CERTIFICATE
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26.11.2025	26.11.2028	2	2

7. Results :

Denomination	Mass Value (g)	Uncertainty (g)
5 kg	5 000.002 1	± 0.002 5
500 g	500.000 14	± 0.000 25
1 mg	0.001 002	± 0.000 003

The reported uncertainty is at coverage factor k = 2 which corresponds to a coverage probability of approximately 95 % for a normal distribution.

8. Date(s) of Calibration : 14.11.2025

9. Remarks : Mass value(s) of the weight(s) is(are) within the maximum errors permissible in E_2 accuracy class of weights as per OIML R 111-1 : 2004.

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